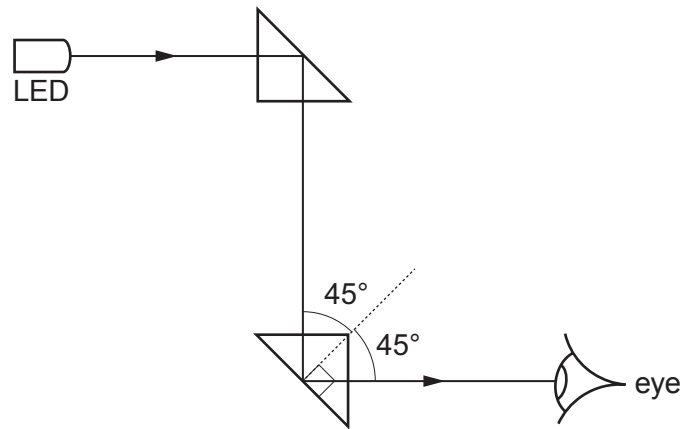


6. (a) A student sets up a simple periscope consisting of two glass prisms ( $n_{\text{glass}} = 1.52$ ). She tests it by using it to view a light emitting diode (LED) as shown.



- (i) Give the full name of the process by which the light changes direction in the prisms. [1]

- (ii) The student wonders whether the periscope will still work properly if the lower prism is surrounded by water ( $n_{\text{water}} = 1.33$ ). Determine whether or not it will, giving your reasoning clearly. [3]

- (b) (i) A particular monomode fibre has a core of refractive index 1.515. Calculate the time it takes for a pulse of light to travel through 1.50 km of the fibre. [3]



- (ii) Explain the advantage of a monomode fibre over a fibre with a much thicker core, for the transmission of a rapid stream of data. [3]

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Examiner  
only

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